Signals & Systems (151-0575-01) Exam, February 7th, 2019 - Scoring rubric (compact, with figures)

Condensed from the official corrected solution. Each criterion heading gives the label, name and maximum. Context is stated once per problem; 'Solution' is the official reference answer including the official point allocation and grading remarks. All answers must be justified. 55 points in total.

Problem 1 context: The causal continuous-time scalar system dy/dt(t) = a y(t) + u(t) is discretized using the forward Euler method for a zero-order hold input and a uniformly sampled output, with sampling time Ts.

1.1.a - Stable sampling times - 3 points:

Question: Determine the values of Ts as a function of a that make the discrete-time system stable.

Solution: Euler: y[n+1] = (1 + a Ts) y[n] + Ts u[n]; stability requires |1 + a Ts| < 1 (1 point). a > 0: always unstable, since Ts > 0 gives 1 + a Ts > 1 (1 point). a = 0: stable for all Ts (1 point). a < 0: stable if Ts < 2/|a| (1 point).

1.2.a - Euler equals exact - 3 points:

Question: Determine for which values of a Euler discretization and exact discretization correspond.

Solution: Build M = [[a, 1], [0, 0]]. Correspondence requires I + Ts A = sum_k (M Ts)^k / k! (1 point), i.e. M^2 = [[a^2, a], [0, 0]] = 0 (1 point), which holds only for a = 0 (1 point).

2.a - System properties - 6 points:

Question: Consider the discrete-time system y[n] = a y[n-3] + u[n] + 0.5 cos(2 pi n), with a in R. Elaborate on linearity, stability, causality, time-invariance and memorylessness as the parameter a in R varies. Justify your answer.

Solution: Non-linear (affine) for all a (1 point); time-invariant for all a, since cos(2 pi n) = 1 is constant (1 point; a convincing argument for time-variance also gets full marks); memoryless only for a = 0 (1 point); causal and stable for |a| < 1, anti-causal and stable for |a| > 1 (3 points; only 1.5 if the anti-causal case is missing). In general 0.5 points each for a correct conclusion and a correct argument.

Problem 3 context: Consider the cascade of two LTI systems, u[n] -> G_1 -> G_2 -> y[n], with impulse responses h_1[n] = sin(8n) and h_2[n] = a^n s[n], with |a| < 1, where s[n] is the unit step.

3.1.a - Output of G2 - 3 points:

Question: Determine the output of system G_2 if the input is u[n] = delta[n] - a delta[n-1], where delta[n] is the unit impulse.

Solution: u[n] * h_2[n] = a^n s[n] - a^n s[n-1] = delta[n]: the output is a unit impulse.

3.2.a - Output of the cascade - 5 points:

Question: Determine the output of the cascade of G_1 and G_2 if the input is u[n] = delta[n] - a delta[n-1].

Solution: LTI systems commute, so u * (h_2 * h_1) = (u * h_2) * h_1 (3 points). The output is therefore the impulse response of G_1: delta[n] * sin(8n) = sin(8n) (2 points). Small errors: -0.5 or -1 point.

Problem 4 context: Consider the signal x[n] = 0 for n <= 0, x[n] = 0 for n odd and n > 0, x[n] = 1/n^2 for n even and n > 0. Let X(Omega) be the Fourier transform of {x[n]}, Y(Omega) = e^(j Omega) X(Omega), Q(Omega) = X(Omega) + Y(Omega), q[n] the inverse Fourier transform of Q(Omega), and r[n] = 0 for n <= 0, r[n] = 0 for n odd and n > 0, r[n] = q[n] - q[n-2] for n even and n > 0.

4.a - Signal r[n] - 6 points:

Question: What is r[n]?

Solution: sum 1/n^2 converges, so x is absolutely summable and X(Omega) exists (1 point). By linearity q = x + F^-1(e^(j Omega) X) (1 point), and F^-1(e^(j Omega) X) = {x[n+1]}, so q[n] = x[n] + x[n+1] (2 points). q = {..., 0 (n=0), 1/2^2, 1/2^2, 1/4^2, 1/4^2, ...} (1 point). Hence r[2] = 1/2^2 - 0, r[4] = 1/4^2 - 1/2^2, and in general r[n] = 1/n^2 - 1/(n-2)^2 for even n >= 4, with r[n] = 0 for n <= 0 and odd n (1 point; -0.5 if n = 2 is missing).

Problem 5 context: Let x[n] and y[n] be two periodic signals, both with period N. Let y[n] != 0 for all n, and z[n] = sum_{k=0}^{N-1} x[k] / y[n-k].

5.1.a - Periodicity of z[n] - 2 points:

Question: Is the signal z[n] periodic? If yes, what is its period?

Solution: z[n+N] = sum_k x[k] / y[n+N-k] = sum_k x[k] / y[n-k] = z[n], so z is periodic (1 point) with period N (1 point).

5.2.a - DFS of z[n] - 3 points:

Question: Let X[k], W[k], Z[k] be the Fourier (DFS) coefficients of x[n], w[n] = 1/y[n] and z[n], respectively. Express Z[k] as a function of X[k] and W[k].

Solution: Z[k] = 1/N sum_n sum_r x[r]/y[n-r] e^(-j2k pi/N n); splitting the exponential as e^(-j2k pi/N (n-r)) e^(-j2k pi/N r) and reordering the sums gives Z[k] = N X[k] W[k] (1 point per step). Marked rigorously: the steps must be shown; the bare result X[k] W[k] gets 0 points.

6.a - Filter design - 10 points:

Question: A signal u has the form u[n] = c_0 + c_1 (-1)^n + c_2 cos(pi/3 n), n = ..., -1, 0, 1, ..., where c_0, c_1, c_2 in R are fixed but unknown constants. Design a causal, stable, LTI filter G so that if u has the form above and y = G u, then y[n] = c_0 for all n. Hint: use cos(pi/3) = 0.5.

Solution: Write u with frequencies Omega_0 = 0, Omega_1 = pi, Omega_2 = pi/3 (1 point). Need H(1) = 1, H(-1) = 0, H(e^(+-j pi/3)) = 0 (1 point each). FIR guarantees stability (1 point); causality by multiplying by z^-N (1 point). H(z) = alpha (z + 1)(z^2 - z + 1) z^-3 = alpha (1 + z^-3) (1 point); H(1) = 1 gives alpha = 1/2 (1 point), so H(z) = (1 + z^-3)/2 and y[n] = (u[n] + u[n-3])/2 (1 point). An order-6 moving average is also accepted.

Problem 7 context: To identify the coefficients of the transfer function of a causal, stable LTI system of the form H(z) = b_0 / (1 + a_1 z^(-1)), the signal {u[n]} = {(e^(j pi/2 n) + e^(-j pi/2 n)) / 2} is given as input. The Discrete Fourier Series coefficients of the periodic output {y[n]} are {Y[k]} = {0, 3.2 + j1.6, 0, 3.2 - j1.6}. Determine the values of the coefficients a_1 and b_0.

7.a - Identify a1 and b0 - 6 points:

Question: Determine the values of the coefficients a_1 and b_0. (The submission may answer this as two parts, 7a = a_1 and 7b = b_0.)

Solution: DFS of the input: {U[k]} = {0, 2, 0, 2} (2 points). DFS of the impulse response {H[k]} = {Y[k]/U[k]} = {0, 1.6 + j0.8, 0, 1.6 - j0.8} (1 point). Real and imaginary parts give the linear system F [a_1, b_0]^T = G with F = [[-0.8, 1], [1.6, 0]], G = [1.6, 0.8] (2 points). Result: a_1 = 0.5 and b_0 = 2 (1 point). Calculating everything in the time domain is also accepted.

Problem 8 context: For the following pair of signals {u[n]} and {y[n]}, determine whether {y[n]} can be the output of a discrete-time LTI system given the input {u[n]}. If such a system exists, determine whether it is unique, i.e. fully defined by this input-output relationship. If no such system exists, explain why.

8.1.a - Pair 1 complex exponential - 2 points:

Question: {u_1[n]} = {e^(jn)}, {y_1[n]} = {2 e^(jn)}.

Solution: An LTI relationship exists; it only requires H(e^(j)) = 2 (H at Omega = 1), so the system is not unique.

8.2.a - Pair 2 causal cosine - 2 points:

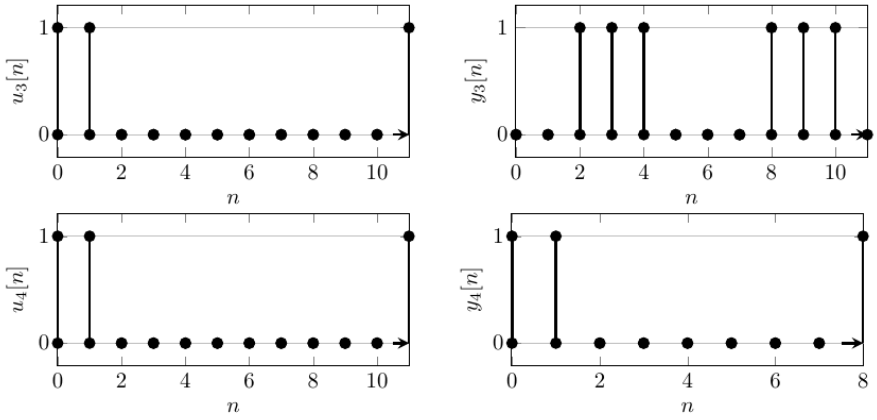
Question: {u_2[n]} = {cos(n) s[n]}, {y_2[n]} = {2 cos(n) s[n]}, where {s[n]} is the unit step.

Solution: An LTI relationship exists and is uniquely defined: H(Omega) = 2 (i.e. y = 2u).

8.3.a - Pair 3 periodic signals - 2 points:

Question: {u_3[n]} and {y_3[n]} are both periodic and depicted in the time domain in a figure (n = 0..10, values between 0 and 1) (see figure below).

Solution: An LTI relationship exists because both signals are periodic with the same fundamental period, but it is not unique: only the values H(e^(j 2pi k/12)) are specified, the rest of H(Omega) is arbitrary.



Signals for pairs 3 (top: u_3, y_3) and 4 (bottom: u_4, y_4), as printed in the exam.

8.4.a - Pair 4 periodic signals - 2 points:

Question: {u_4[n]} and {y_4[n]} are both periodic and depicted in the time domain in a figure (n = 0..10, values between 0 and 1) (see the figure under criterion 8.3.a).

Solution: No LTI relationship exists, because the signals have different periods; this would violate the eigenfunction-eigenvalue property.